

Post-Disaster Rescue Guidance via VLM Fine-Tuning

CSE499A – Section 15, Group 5, Update 1

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Weekly Update:

Our primary objective this week was to architect and construct a highly robust, end-to-end data curation pipeline for our Vision Language Model (VLM). The goal was to programmatically ingest, normalize, and process raw, unstructured disaster imagery into a highly balanced, tactical instruction-response dataset ready for QLoRA fine-tuning.

Work Done by Abrar Mohammed Tanzim Alam

1. **Image Standardization Pipeline:** Built robust pre-processing scripts ([support/image_standardizer.py](#)) using PIL to enforce RGB color spaces, cap max resolution at 1280px (preventing VRAM OOM errors during Qwen2-VL training), filter images below 256px, and output standardized .jpg and .json metadata with resume-safe execution tracking.
2. **Instruction-Response Ground Truth Generation:** Engineered generation engines ([support/generate_ground_truth.py](#), [support/aider_generate_ground_truth.py](#)) via google.genai SDK. Implemented dynamic prompt multipliers to over-sample rare events, yielding a balanced pilot batch of 496 pairs for tactical rescue guidance ([dataset/train_dataset.jsonl](#)).
3. **Dataset Sanitization Engine:** Created an automated regex-based sanitization script ([support/sanitize_dataset.py](#)) to scrub conversational AI language, enforce strict length boundaries (80-600 words), and inject 6 diverse tactical instruction variants via round-robin, outputting the randomized [dataset/final_training_dataset.jsonl](#).
4. **Final Verified Dataset Statistics Visualization:** Developed [support/dataset_stats.py](#) to parse the finetune-ready dataset and plot post-generation distributions via `matplotlib/seaborn`, proving our balancing mechanisms mitigated intrinsic biases.
5. **Cross-Dataset Pipeline Adaptation (AIDER Transfer):** Refactored the standardizer and generation scripts to dynamically interpret AIDER's unique class-based structure and normalize coordinates without disrupting the core pipeline, enabling AIDER data to utilize xView's downstream processing.
6. **Synthetic Metadata Synthesization:** Designed a bridging script ([support/aider_synthesize_metadata.py](#)) to query Gemini for synthetic xBD-equivalent metadata (polygons and classifications) to normalize AIDER's lack of ground-truth annotations into our strict uniform training schema.

Joint Contribution: T. Mou, A. Sami, and R. Afrin

1. **Validation UI Tool:** Developed a Tkinter GUI ([support/dataset_validator.py](#)) for rapid verification. Ingests raw data to allow Accept/Reject evaluation against a Rescue Actionability Rubric (RAR), checking for collapses, mapping, and hazard avoidance, outputting verified samples.

Work Done by Ridita Afrin Riya

1. **AIDER Adaptation & Processing:** Adapted the standardization script ([support/aider_image_standardizer.py](#)) for AIDER and processed half (~2700) of the raw dataset. Generated 50 instruction-response pairs.

Work Done by Aryan Sami

1. **Validation & Processing:** Primary reviewer for 200 xView pairs via the GUI validator. Processed the remaining half (~2700) of AIDER and generated 50 instruction-response pairs.

Work Done by Tamanna Akter Mou

1. **Exploratory Data Analysis (EDA):** Analyzed AIDER and xView distributions (resolutions, categories, severities) via Jupyter notebooks, exposing imbalances that guided the pipeline's architecture.

Work Done by Abdullah Al Noman

1. **Validation & Generation:** Secondary reviewer for 300 xView pairs against the strict RAR. Generated 50 AIDER instruction-response pairs.

Next Week's Plan: We will proceed to merge the fully processed datasets and begin the QLoRA fine-tuning phase using the prepared splits to train the Qwen2-VL model.